

James Iestyn Rigsby Harris

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🐙 github.com/jamesharris1307 | 🌐 jamesharris.dev | 📍 Cardiff, UK

TECHNICAL EXPERIENCE

Quantum Software Probabilistic Testing Framework

September. 2025 – May. 2026

Probabilistic Testing Framework for Qiskit Aer Simulation

Cardiff, UK

- **Tech Stack** - Python, Docker, Pytest, Qiskit, Qiskit Aer
- Achieved a scalable bridge between classical engineering and quantum research by integrating the testing framework with the Pytest lifecycle, enabling automated verification of quantum properties like Superposition and Entanglement within modern Continuous Integration pipelines.
- Contributed to solving the Test Oracle Problem in quantum computing by replacing deterministic binary pass/fail logic with Threshold Assertions, which allow developers to define statistical tolerance parameters and confidence intervals for probabilistic quantum state distributions.

Parallel Systems Simulation Comparison (Langton's Ant)

October. 2025 – May. 2026

Langton's Ant Cellular Automaton

Cardiff, UK

- **Tech Stack** - Java, JavaFX, Multi-threading, Parallelization
- Engineered a high-concurrency simulation engine using the Strategy Design Pattern, achieving a peak 7.26x speed-up by transitioning from sequential execution to a coarse-grained ExecutorService.
- Optimized memory efficiency by implementing a sparse grid via ConcurrentHashMap and bit-shifting for 64-bit coordinate encoding, reducing space complexity from $O(n^2)$ to $O(K)$, where K represents the active cell count.
- Analysed three distinct concurrency models: Fine-Grained ExecutorService (identifying a 2.3x performance degradation due to task overhead), Coarse-Grained Batching (7.26x speed-up using Recursive Decomposition), and ForkJoinPool (7.18x speed-up using Work-Stealing)

Smart Campus Navigator Application

September. 2024 – May. 2025

Full-Stack Campus Navigation Application

Cardiff, UK

- **Tech Stack** - JavaScript, Node.js, Express.js, MongoDB, Mongoose, Leaflet, HTML/CSS, Docker, Jest
- Developed a full-stack campus navigation system for Cardiff Metropolitan University as part of a three-person development team. Built an interactive Leaflet-based map with searchable campus locations, building information, points of interest, and real-time directions.
- Implemented a Node.js/Express.js REST API with MongoDB and Mongoose for managing application data, alongside authentication and secure password hashing. Used Docker for containerised development and Jest for automated testing.

ACADEMIC EDUCATION

Cardiff Metropolitan University

September. 2023 – May. 2026

BSc (Hons) Software Engineering – First Class Honours

Cardiff, UK

- **Awards** – Awarded **Software Engineering Student of the Year** for Best Overall Performance in the final year and **Progression Award** in the second year.
- **Accreditation** – BCS and IET accredited programme (Partial CEng)

Cardiff and Vale College

September. 2020 – May. 2022

Foundation Degree Graphic Communication – Merit

Cardiff, UK

- **Student Representative**: Represented students in meetings with faculty, academic staff and college management by communicating feedback, concerns, and suggestions to support improvements to both the student and teacher experience.

WORK EXPERIENCE

Fisherman's Rest Inn

July 2021 – September 2024

Supervisor / Front of House

Bedwas, UK

- Led shifts, trained new staff, resolved customer complaints and concerns, managed daily operations such as coordinating staff roles and filling in for missing team members.
- Took delivery orders and bookings over the phone, managed reservations, and provided customer service as a bartender and waiter. Opened and closed the pub, including setting up for service and completing end-of-day tasks.

McDonald's

December 2022 – May 2023

Crew Member

Caerphilly, UK

- Provided friendly and efficient customer service in a fast-paced environment, taking customer orders and communicating clearly through the drive-thru headset.
- Served customers at the drive-thru window, prepared beverages, assembled and bagged orders, and ensured orders were accurate before being handed to customers.
- Maintained high standards of cleanliness throughout the restaurant and worked as part of a team to keep service running efficiently during busy periods.

One Stop Pest Control

May 2022 – June 2022

Pest Control Operative

Bedwas, UK

- Worked as part of a team to sterilise and clean pest-infested areas to ensure they were safe and hygienic following treatment.
- Safely removed contaminated insulation and other hazardous materials from pest-infested spaces.
- Disposed of contaminated and hazardous materials safely and in accordance with relevant health, safety and environmental regulations.

ADDITIONAL INFORMATION

Full UK Driving Licence

UK